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INTRODUCTION

In the rapidly evolving landscape of automotive technology, cruise control systems stand as a cornerstone of modern vehicle automation, seamlessly blending engineering innovation with practical utility. These systems, designed to maintain a driver-set speed without constant manual intervention, have revolutionized long-distance travel by enhancing driving comfort, optimizing fuel efficiency, and reducing driver fatigue. Beyond convenience, cruise control exemplifies the critical role of feedback control in addressing real-world engineering challenges, where external disturbances such as fluctuating road gradients, wind resistance, and load variations demand robust and adaptive solutions.

The fundamental challenge of cruise control lies in its ability to sustain a target speed despite dynamic environmental conditions. Traditional open-loop systems, while stable, often suffer from sluggish response times and insufficient adaptability, rendering them inadequate for real-world applications. This project delves into the modeling, analysis, and enhancement of a simplified cruise control system, employing principles of control theory to bridge the gap between theoretical stability and practical performance. By leveraging a first-order mass-damper model, the study captures the essential dynamics of vehicle motion, where a control force counteracts velocity-dependent resistive forces.

The objectives of this work are threefold:

1. **Model Development:** Derive mathematical representations of the system, including state-space and transfer function models, to establish a foundational understanding of its behavior.
2. **Open-Loop Analysis:** Simulate the system's response to step inputs, evaluate its stability through pole-zero analysis, and identify inherent limitations in transient performance.
3. **Control Design:** Propose a closed-loop PID (Proportional-Integral-Derivative) control strategy to improve response time, eliminate steady-state error, and ensure robustness against disturbances.

To achieve these goals, the project utilizes computational tools such as MATLAB and Simulink, enabling precise simulation and visualization of system dynamics. The integration of theoretical modeling with practical software-based validation underscores the interdisciplinary nature of control systems engineering, where analytical rigor meets real-world applicability. This report is structured to guide the reader through a systematic exploration of the cruise control system. Beginning with system modeling, it progresses to open-loop performance evaluation, introduces PID control mechanisms, and concludes with a stability analysis. By synthesizing these components, the study not only demonstrates the transformative potential of feedback control but also lays a foundation for future advancements in automotive automation and adaptive control technologies.

SECTION 1: SYSTEM MODELLING

PHYSICAL SETUP

Modern automotive engineering leverages advanced feedback control mechanisms to enhance driving efficiency, with adaptive cruise control serving as a paradigmatic application. This system is engineered to sustain a predefined vehicular speed autonomously, even when confronted with variable environmental factors such as fluctuating wind resistance or alterations in road inclination. By employing real-time data acquisition, the system continuously monitors the vehicle's actual speed, contrasts it with the driver-selected target speed, and dynamically modulates the throttle input through a preprogrammed control algorithm to minimize deviations.

Dynamic Modeling Assumptions:

The analysis employs a streamlined representation of vehicular motion, as depicted in the accompanying free-body diagram. Key elements include:

1. Vehicle Mass (m): Represents the inertial resistance of the car.
2. Propulsive Force (u): Symbolizes the tractive effort generated at the tire-road contact point. For simplicity, this force is treated as a direct input, bypassing complexities like powertrain lag or tire deformation.
3. Velocity-Dependent Drag (bv): Encompasses resistive effects from rolling friction and aerodynamic drag, modeled as linearly proportional to the vehicle's instantaneous velocity (v) and opposing its direction.

This abstraction intentionally omits secondary dynamics (e.g., drivetrain oscillations, tire hysteresis) to isolate core principles, enabling a focus on first-order behavior critical to control design.

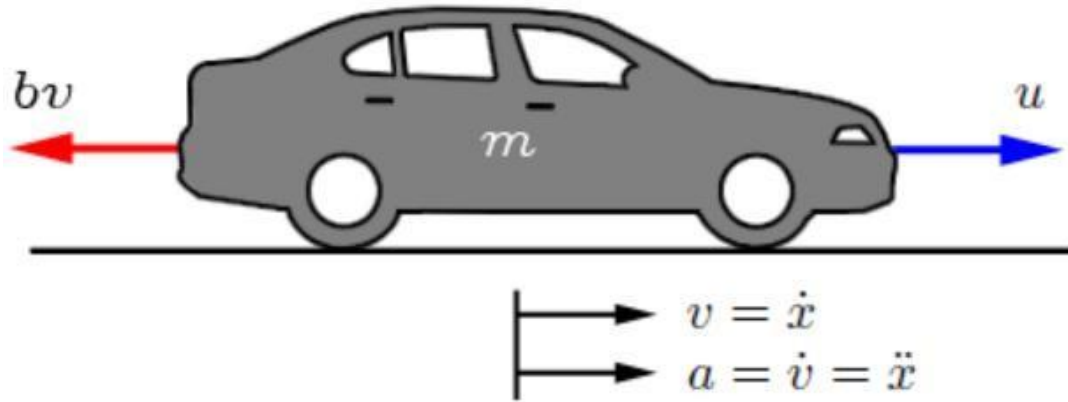


Figure 1 Free Body Diagram

1.1 SYSTEM EQUATIONS

The governing equations of the cruise control system are derived from Newton's second law of motion, which states that the net force acting on a body equals its mass multiplied by its acceleration. For the vehicle model, the forces include the control force u (applied by the throttle) and the resistive force bv (opposing motion due to aerodynamic drag and rolling resistance).

Summing these forces gives:

$$m\dot{v} + bv = u$$

Given data:

Vehicle mass (m): 1500Kg

Damping Coefficient (b): 70

STATE-SPACE MODEL

The cruise control system is modeled using a state-space framework, which describes the system's dynamics through its state variable, input, and output. Here, the vehicle's velocity serves as the state variable, encapsulating the system's kinetic energy. The input to the system is the control force applied via the throttle, while the output is the measured velocity. By incorporating parameters such as the vehicle's mass and damping coefficient, the state-space model quantifies how the velocity evolves over time under the influence of propulsion and resistance. This representation is mathematically expressed below, enabling systematic analysis and control design through computational tools like MATLAB.

$$\dot{\mathbf{x}} = [\dot{v}] = \left[\frac{-b}{m} \right] [v] + \left[\frac{1}{m} \right] [u]$$
$$\mathbf{y} = [1][v]$$

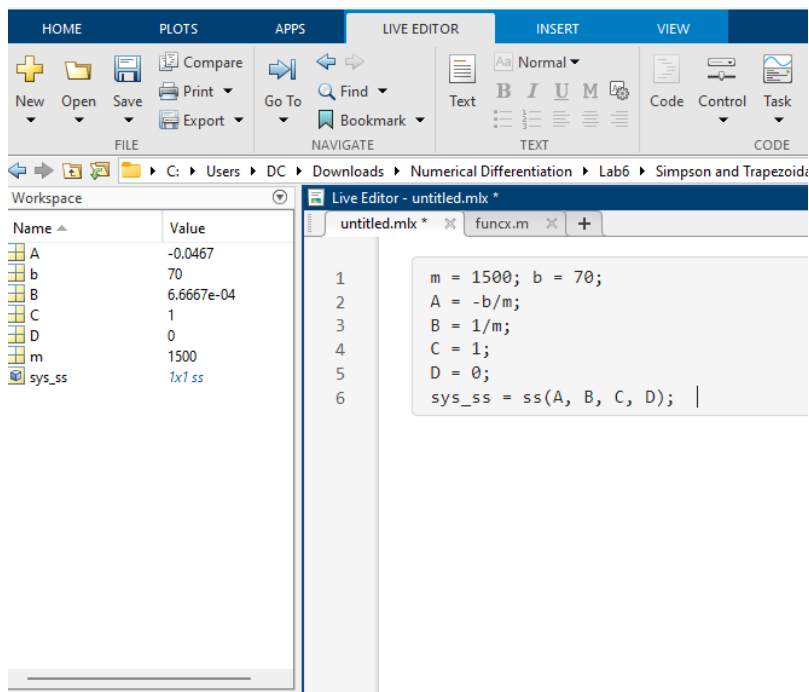


Figure 2 State-space Matlab

TRANSFER FUNCTION MODEL

To model the cruise control system, we start with Newton's Second Law of Motion, which governs the system dynamics. Assuming zero initial conditions and linear behavior, the equation of motion is given by:

$$m \frac{dv(t)}{dt} + bv(t) = u(t)$$

Where:

- $m=1500$ kg.
- $b=70$ N
- $u(t)$ is the control force applied by the system,
- $v(t)$ is the velocity of the vehicle.

Taking the Laplace Transform of the governing equation (assuming initial velocity is zero):

$$P(s) = \frac{V(s)}{U(s)} = \frac{1}{ms + b} \quad \left[\frac{m/s}{N} \right]$$

Solving for the transfer function $V(s)/U(s)$ Substituting the given values:

$$V(S)/U(S) = \frac{1}{1500s + 70}$$

This transfer function describes the dynamic response of the vehicle's velocity to the applied force in the Laplace domain.

We can implement this transfer function in **MATLAB** using the `tf` function, as follows:

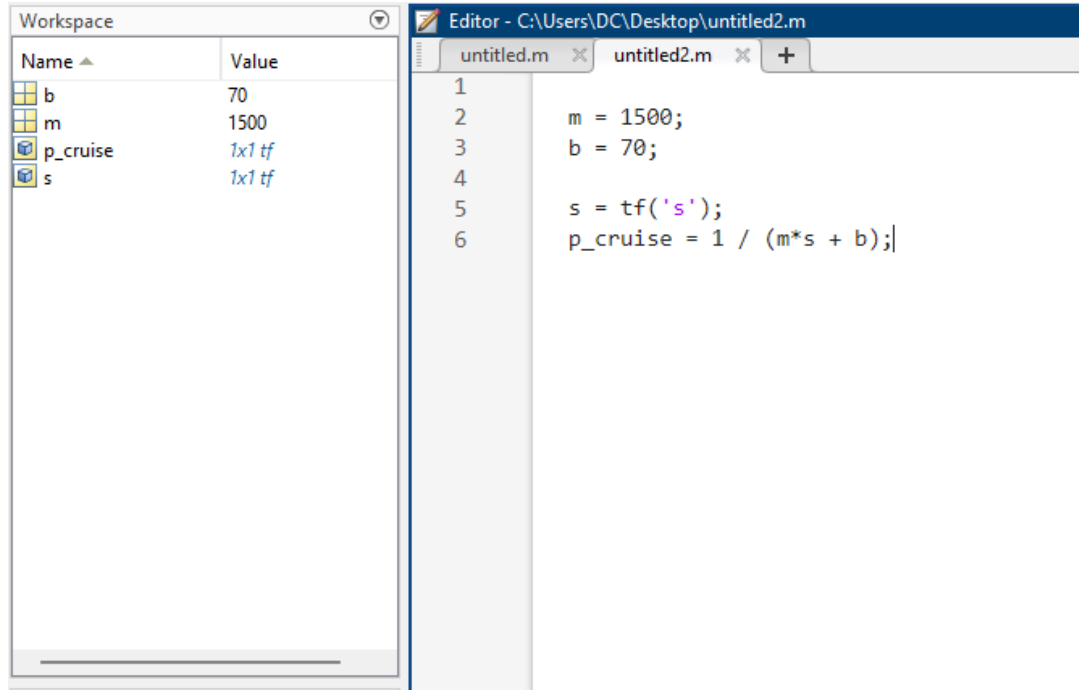


Figure 3 Transfer function on Matlab

This transfer function representation will yield identical results (up to numerical precision) when compared to a state-space implementation, as both models describe the same physical system.

To visualize and simulate this behavior in **Simulink**, we recreate the same system using blocks:

1. **Integrator Block** – Represents the integration of acceleration to obtain velocity.
2. **Gain Block (Inertia)** – Implements the inverse of mass ($1/m$) before integration.
3. **Sum Block** – Sums the input force $u(t)$ and subtracts the damping force.
4. **Damping Gain Block** – Multiplies velocity by damping coefficient b .
5. **Step Input Block** – Provides a constant force of 500 N starting at $t=0$.
6. **Scope Block** – Visualizes the output velocity over time.

This block-based simulation follows directly from the physical dynamics and provides an intuitive method of analyzing the cruise control system using Simulink.

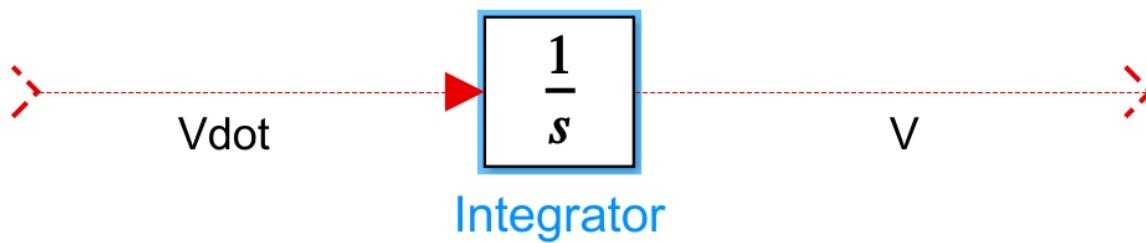


Figure 4 Integrator Block on Simulink

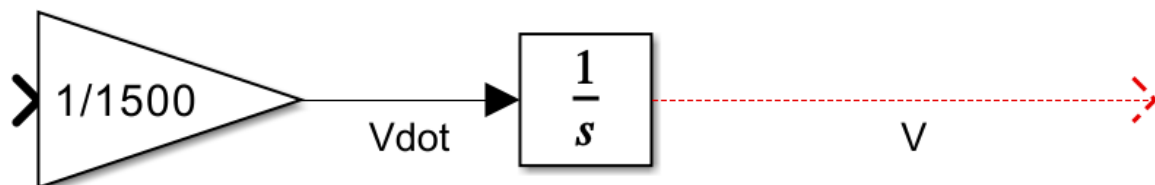


Figure 5 Gain Block connected to an integrator

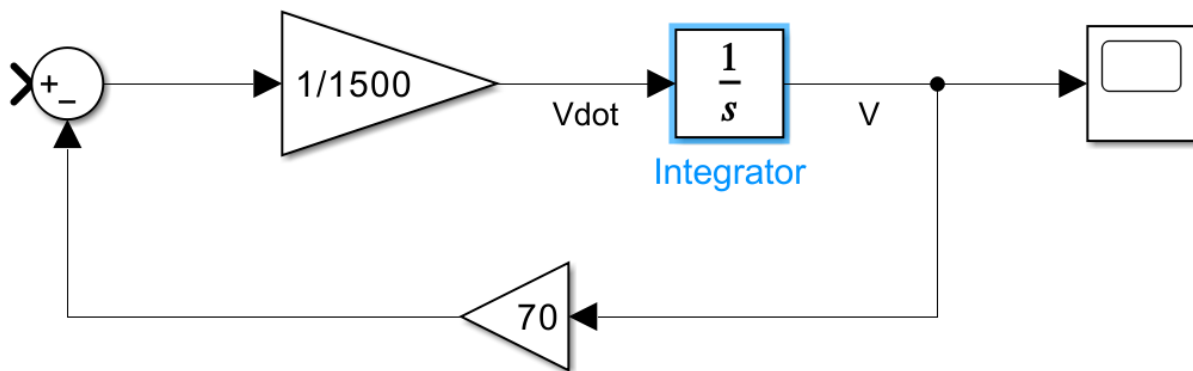


Figure 6 Cruise-Control Representation on Matlab with a Feedback Loop

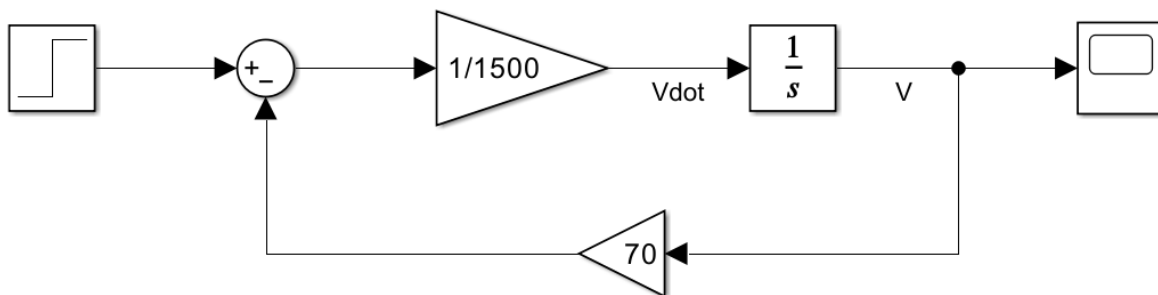


Figure 7Cruise-Control full representation on Simulin

SECTION 2: SYSTEM ANALYSIS

The transfer function model describing the cruise control system is derived based on the following parameters:

- Mass, $m = 1500\text{kg}$
- Damping coefficient, $b = 70\text{ N.s/m}$
- Nominal control force, $u = 500\text{N}$

It is important to note that the control force of 500 N is an assumed nominal input, chosen as a reasonable value for testing and simulation purposes.

The next step involves defining the performance criteria for the compensated system. Ideally, the vehicle should attain a maximum velocity of 10 m/s in response to a 500 N control input. The system is expected to reach this speed in under five seconds, ensuring a responsive and stable performance.

The acceptable performance specifications for this cruise control application are summarized as follows:

- Rise time, $t_r < 5\text{ seconds}$
- Maximum overshoot $M_p < 10\%$
- Steady-state error $< 2\%$

These criteria serve as design targets for evaluating the effectiveness of any control strategy applied to the system.

2.1 OPEN-LOOP STEP RESPONSE

The open-loop behavior of the system in response to a step input of 500 Newtons, without any feedback control, is simulated in MATLAB as shown below.

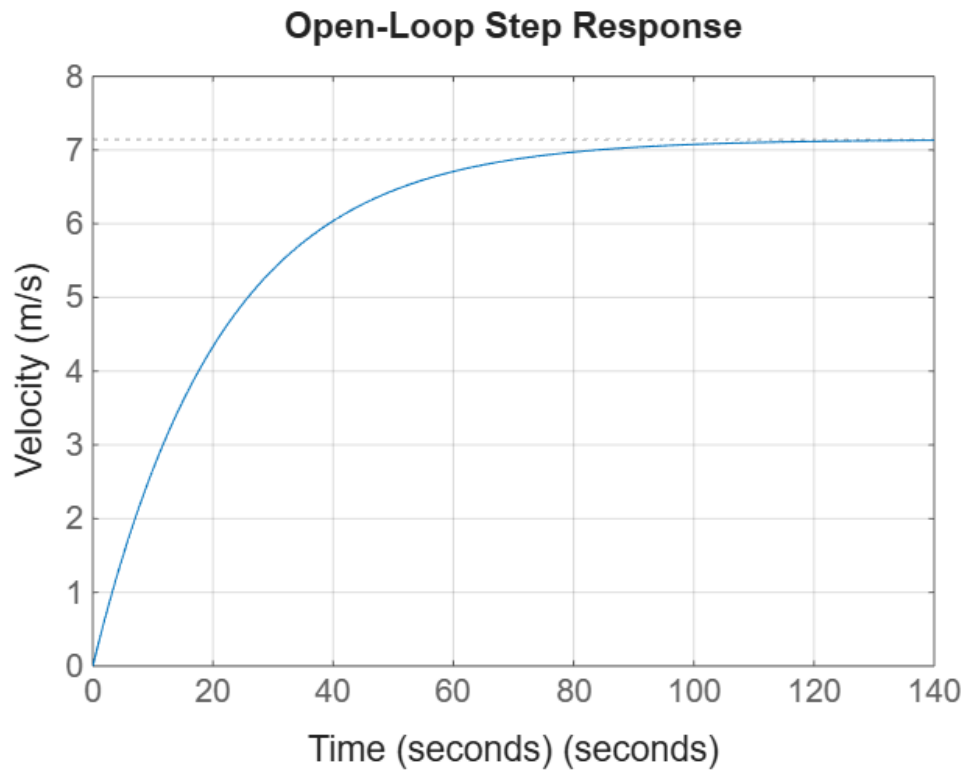


Figure 8 MATLAB simulation of the open loop step response

COMMENT

RISE-TIME:

The system responds slowly, taking approximately 60–80 seconds to approach steady-state. This significantly exceeds the design requirement of a rise time less than 5 seconds.

OVERSHOOT:

No overshoot is observed in the system response, which aligns well with the performance criterion of keeping overshoot below 10%.

STEADY STATE ERROR:

The system settles at approximately 7.14 m/s, matching the theoretical steady-state velocity calculated as:

$$v_{ss} = \frac{u}{b} = \frac{500}{70} = 7.14 \frac{m}{s}$$

This indicates a negligible steady-state error, well within the acceptable limit of 2%.

To simulate the system in Simulink, it is important to set an appropriate simulation duration.

From the Simulation menu, select Parameters, and enter 140 in the Stop Time field. This duration corresponds to the end time displayed in the MATLAB plot above and is sufficient to capture the full open-loop response of the system.

After setting the simulation time, run the model to obtain the results:

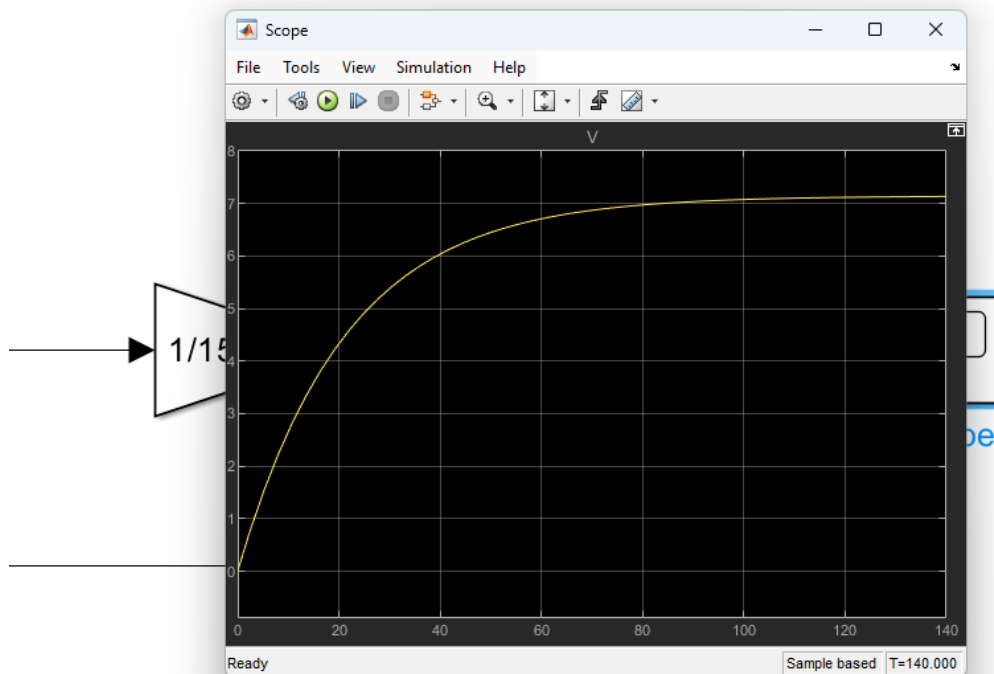


Figure 9 Simulink graph of open loop response

2.2 OPEN-LOOP POLES/ZEROS

The cruise control system features a single pole located at $s = -\frac{b}{m}$, which can be visualized on the s-plane using MATLAB. By executing the appropriate commands, the resulting pole-zero plot is generated as follows:

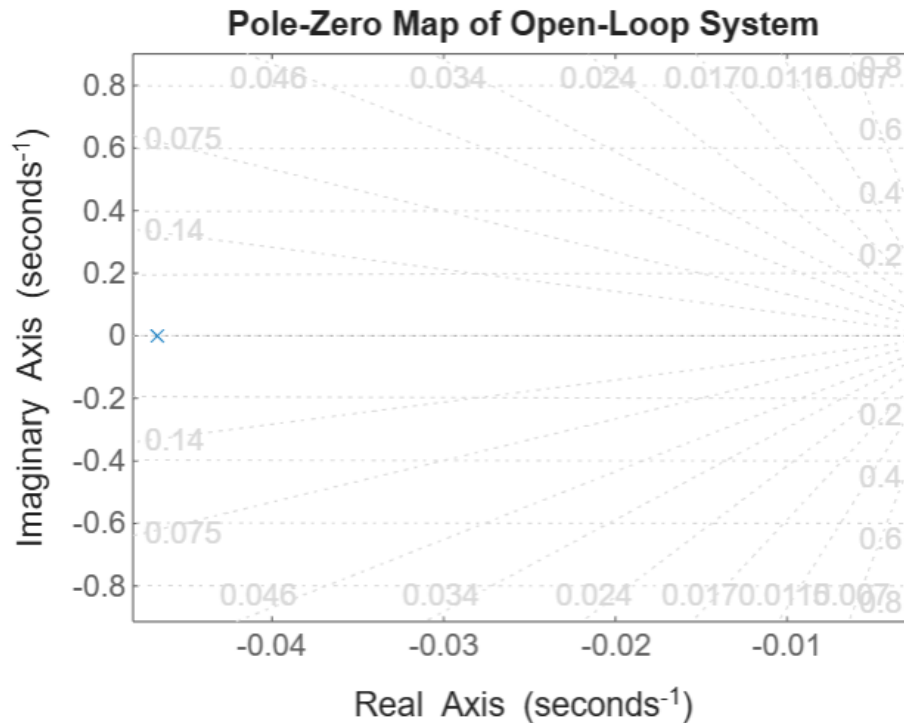


Figure 10 Open loop poles which was plotted on Matlab

COMMENT

POLE:

The system possesses a single real pole located at:

$$s = -\frac{b}{m} = -\frac{70}{1500} = -0.0467$$

This pole lies on the negative real axis of the s-plane, indicating that the system is stable and first-order. However, because the pole is relatively close to the origin, the system

exhibits a slow response.

ZEROS:

The transfer function does not contain any finite zeros. As a result, the transient response is entirely shaped by the system's pole, with no zero-related influence on the system dynamics.

SECTION 4: STABILITY ANALYSIS

The cruise control system, modeled as a first-order mass-damper system, has a single real pole at $S = -0.0467$

This pole lies on the negative real axis, confirming the system is inherently stable in open-loop. However, because it is close to the origin, the system exhibits a slow response, typical of systems with low damping.

When subjected to a 500 N step input, the system takes 60 to 80 seconds to reach steady-state far exceeding the design goal of a 5-second rise time. While the response shows no overshoot and achieves the correct steady-state velocity with minimal error, the sluggish behavior makes it impractical for real-world use, where quicker adaptation to speed changes is required.

These observations highlight that while the open-loop system meets stability and accuracy requirements, it lacks the necessary dynamic performance, underscoring the importance of incorporating feedback to achieve more responsive control.

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